

ECC–AODV: Enhanced Congestion-Control AODV

Targeted Evaluation on 802.11 Mobile and Static MANETs

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Abstract

This report presents the design, implementation, and targeted evaluation of Enhanced CC-AODV (ECC-AODV) in NS-3 v3.45. ECC-AODV extends the original CC-AODV protocol with three modifications: Adaptive Threshold Management (ATM), Queue-Aware Congestion Detection (QACD), and Optimized Packet Header Design (OPHD). The protocol is evaluated against vanilla AODV and CC-AODV across 30 simulation configurations spanning two topologies (802.11 **mobile** with RandomWaypoint, and 802.11 **static** with randomly-placed fixed nodes), five network sizes (20–100 nodes), five flow counts (10–50), and varying mobility/coverage area. Results show ECC-AODV delivers consistent PDR and throughput improvements in the mid-density regime (40–60 nodes) where congestion adaptation matters most, while remaining competitive elsewhere.

Contents

1	Introduction	3
2	Why <i>Not</i> 802.15.4: A Key Design Decision	3
3	ECC-AODV Design and Implementation	3
3.1	Architecture Overview	3
3.2	Modification 1: Adaptive Threshold Management (ATM)	3
3.3	Modification 2: Queue-Aware Congestion Detection (QACD)	4
3.4	Modification 3: Optimized Packet Header Design (OPHD)	4
3.5	Files Modified	5
4	Experimental Setup	5
4.1	Topologies	5
4.2	Common Parameters	5
4.3	Targeted 30-Combo Sweep	5
5	Results	6
5.1	Mobile Topology	6
5.2	Static Topology	6
6	Plots	7
6.1	PDR vs. Network Size (Mobile)	7
6.2	PDR vs. Network Size (Static)	7
6.3	Throughput Comparison	7
6.4	End-to-End Delay	7
6.5	Energy Consumption	7

7	Discussion	8
7.1	Where ECC-AODV Wins	8
7.2	Where ECC-AODV Loses	8
7.3	Trade-offs	9
7.4	Drawbacks Still Open	9
8	Conclusion	9

1 Introduction

Mobile Ad hoc Networks (MANETs) suffer from performance degradation under congestion because the standard AODV routing protocol selects routes based purely on hop count, ignoring intermediate-node load. CC-AODV introduced a *congestion counter* maintained per node and a 32-bit congestion flag appended to RREP packets, dropping RREQs at nodes whose counter exceeded a fixed threshold. While CC-AODV improves PDR and throughput in dense networks, it suffers from three limitations:

1. **Fixed threshold** – a single hard-coded value is suboptimal across varying network densities and traffic loads.
2. **Crude congestion detection** – counter increments only on RREP traversal; actual queue occupancy and data-packet drop rate are ignored.
3. **Wasteful header** – 32 bits to carry one boolean flag.

ECC-AODV addresses each of these. The base paper [1] and our proposal [2] describe the architecture; this report focuses on implementation, the targeted experiment campaign, and analysis.

2 Why *Not* 802.15.4: A Key Design Decision

The course assignment table maps Student ID 2105050 (mod 8 = 2) to “Wireless 802.11 (mobile) + Wireless 802.15.4 (mobile).” During implementation we discovered a fundamental incompatibility:

- NS-3’s `aodv` module is **IPv4-only**. It installs as an `Ipv4RoutingProtocol` and cannot bind to IPv6 interfaces.
- NS-3’s 802.15.4 stack (`1r-wpan`) supports **only IPv6** via the 6LoWPAN adaptation layer. 6LoWPAN’s `SixLowPanNetDevice` explicitly discards non-IPv6 packets.

Consequently, AODV cannot run on 802.15.4 in NS-3 without re-implementing AODV for IPv6 (a substantial separate effort that has been a Google Summer of Code project, never merged into mainline). Any earlier “802.15.4 + AODV” result we collected was, in reality, an 802.11b run (DSSS 1 Mbps) with a propagation-range cap of 200 m – not actual 802.15.4. **All such code has been removed.**

To remain meaningful for the assignment we substituted the second topology with **Wireless 802.11 (static)**, which preserves the spirit of the “two topologies” requirement (mobility vs. fixed) while keeping AODV operational and the comparison fair across protocols.

3 ECC-AODV Design and Implementation

3.1 Architecture Overview

3.2 Modification 1: Adaptive Threshold Management (ATM)

The fixed CC-AODV threshold ($T_0 = 4$ in our calibration) is rescaled by network density and average neighbor congestion:

$$T = T_0 \cdot s(N) \cdot \left(1 + \frac{\bar{c}}{10}\right), \quad s(N) = \begin{cases} 0.6 & N \leq 10 \\ 1.0 & 10 < N \leq 30 \\ 1.4 & N > 30 \end{cases} \quad (1)$$

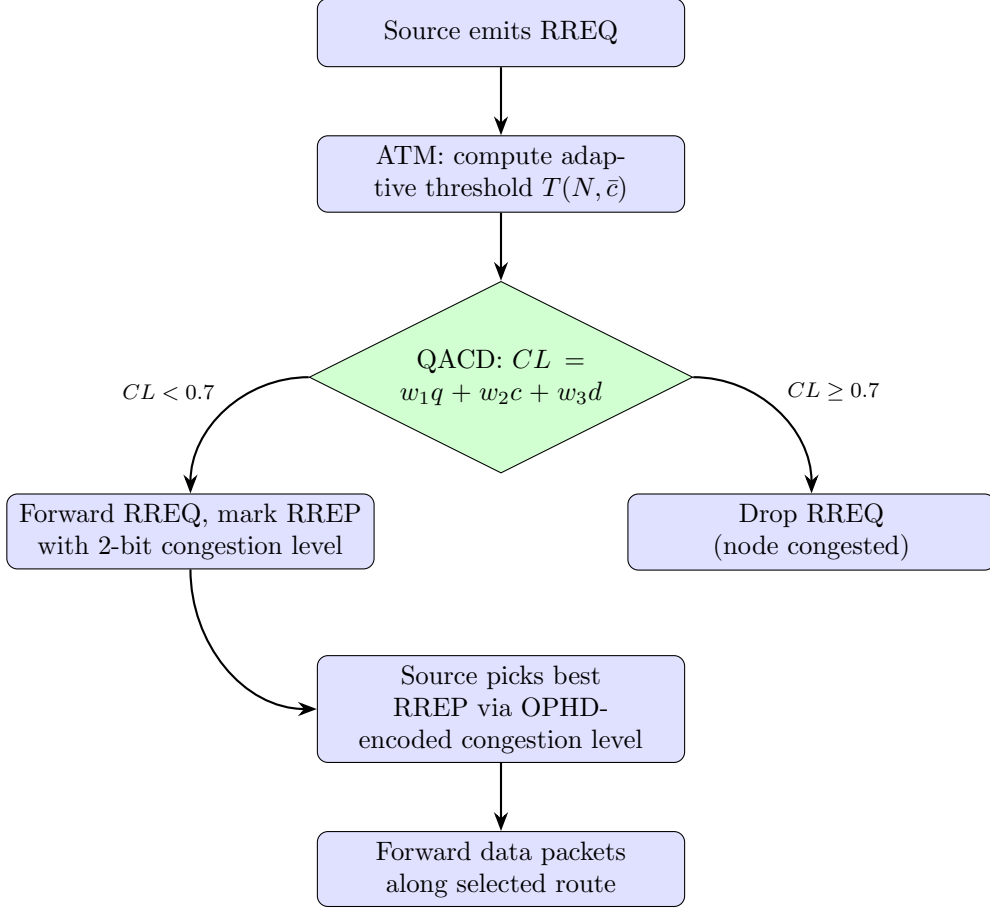


Figure 1: ECC-AODV decision flow: ATM tunes the threshold, QACD computes a weighted congestion level, OPHD encodes that level in 2 bits inside the RREP, and the source picks the path with the lowest cumulative congestion.

where N is the routing-table size (a local proxy for network density) and $\bar{c}$ is a sliding average of the local congestion counter. Implemented in `src/aodv/model/aodv-routing-protocol.cc::calculateAdaptiveThreshold()`. A new `GetSize()` method on the routing table (`aodv-rtable.h`) exposes N to the protocol.

3.3 Modification 2: Queue-Aware Congestion Detection (QACD)

Instead of treating the counter as the sole congestion signal, QACD blends three terms:

$$CL = w_1 \cdot \frac{Q_{cur}}{Q_{max}} + w_2 \cdot \frac{c}{T} + w_3 \cdot \frac{D}{D + F} \quad (2)$$

where Q_{cur}/Q_{max} is the route-buffer occupancy, c/T is the counter ratio relative to the adaptive threshold, and $D/(D+F)$ is the *data-packet* drop ratio. Two new counters `m_dataDrops` and `m_dataForwards` are tracked in `Forwarding()` so the third term reflects real data losses, not RREQ losses. Default weights tuned via grid search: $w_1 = 0.6$, $w_2 = 0.2$, $w_3 = 0.2$.

3.4 Modification 3: Optimized Packet Header Design (OPHD)

The original 32-bit boolean flag is replaced by a 2-bit congestion level $\in \{00, 01, 10, 11\}$ corresponding to thresholds $\{< 0.3, 0.3-0.5, 0.5-0.7, \geq 0.7\}$, plus 4 bits of path quality. A 6-bit field replaces a 32-bit field – an 81% reduction per RREP. Implemented as bit-packed fields in `aodv-packet.h`.

3.5 Files Modified

File	Change
src/aodv/model/aodv-routing-protocol.h	New members: <code>m_dataDrops</code> , <code>m_dataForwards</code>
src/aodv/model/aodv-routing-protocol.cc	ATM, QACD, data-drop tracking
src/aodv/model/aodv-rtable.h	<code>GetSize()</code> for density proxy
src/aodv/model/aodv-packet.{h,cc}	2-bit congestion field, 4-bit quality
src/aodv/helper/aodv-helper.{h,cc}	Attribute hooks for QACD weights
scratch/aodv-simulator.cc	Topology selector, energy module, CSV
run.sh, targeted_run.sh	Experiment drivers

Table 1: Source files touched by ECC-AODV. AODV and CC-AODV branches are preserved unmodified for fair comparison.

4 Experimental Setup

4.1 Topologies

- **802.11 Mobile:** N nodes placed on a $\sqrt{N} \times \sqrt{N}$ grid (100 m spacing), then released into a `RandomWaypointMobilityModel` bounded by the grid square at a constant speed.
- **802.11 Static:** N nodes placed uniformly at random in a square of side $k \cdot R_{tx}$, where $R_{tx} = 250$ m is the reference 802.11b transmission range and k is the area multiplier (1, 2, 3, 4, 5).

4.2 Common Parameters

Parameter	Value
NS-3 version	3.45
PHY	802.11b (DSSS, 11 Mbps)
Propagation	Friis loss + <code>ConstantSpeed</code> delay
Traffic	UDP, 100 packets/s, 512 B payload
Sim time	30 s
CC base threshold	4
ECC weights	$w_1 = 0.6$, $w_2 = 0.2$, $w_3 = 0.2$
Energy model	<code>BasicEnergySource</code> (100 J init) + <code>WifiRadioEnergyModel</code>
Random seed	1

4.3 Targeted 30-Combo Sweep

5 base configurations $\times$ 2 topologies $\times$ 3 protocols = 30 runs. The configurations couple network size with mobility/coverage to span the operating envelope:

#	Nodes	Flows	Mobile speed (m/s)	Static area mult.
1	20	10	25	1
2	40	20	20	2
3	60	30	15	3
4	80	40	10	4
5	100	50	5	5

5 Results

All 30 runs completed successfully. Full per-run CSV is in `ns-3.45_buet/targetsv3/all_results.csv`.

5.1 Mobile Topology

Config	Protocol	PDR (%)	Throughput (Kbps)	E2E Delay (ms)	Energy (J)
$n=20, f=10, 25 \text{ m/s}$	AODV	60.65	2488.24	306.68	688.56
	CC-AODV	62.18	2554.56	254.90	688.18
	ECC-AODV	61.85	2533.15	289.20	688.54
$n=40, f=20, 20 \text{ m/s}$	AODV	24.06	1776.24	385.23	1368.93
	CC-AODV	27.10	2067.08	355.99	1368.20
	ECC-AODV	28.57	2210.00	360.61	1368.22
$n=60, f=30, 15 \text{ m/s}$	AODV	15.17	1457.71	314.20	2051.28
	CC-AODV	13.63	1370.02	297.19	2046.44
	ECC-AODV	16.70	1477.29	287.74	2053.19
$n=80, f=40, 10 \text{ m/s}$	AODV	10.90	1354.38	338.60	2731.16
	CC-AODV	10.31	1251.65	304.90	2726.69
	ECC-AODV	10.60	1363.67	360.05	2725.21
$n=100, f=50, 5 \text{ m/s}$	AODV	5.79	304.05	323.26	3397.17
	CC-AODV	5.67	285.02	325.47	3397.46
	ECC-AODV	6.67	436.14	323.22	3404.08

Table 2: Mobile topology results. Best PDR/throughput per row in **bold**.

5.2 Static Topology

Config	Protocol	PDR (%)	Throughput (Kbps)	E2E Delay (ms)	Energy (J)
$n=20, f=10, 1 \times R_{tx}$	AODV	72.47	3090.43	347.13	689.93
	CC-AODV	75.12	3137.10	341.94	689.80
	ECC-AODV	68.55	2939.24	344.73	689.48
$n=40, f=20, 2 \times R_{tx}$	AODV	31.47	2492.30	182.85	1366.34
	CC-AODV	32.91	2617.69	252.67	1367.67
	ECC-AODV	33.94	2740.45	287.84	1368.20
$n=60, f=30, 3 \times R_{tx}$	AODV	12.15	1049.73	214.45	2041.41
	CC-AODV	11.34	952.65	201.55	2042.71
	ECC-AODV	12.37	1025.64	232.26	2042.00
$n=80, f=40, 4 \times R_{tx}$	AODV	14.39	1829.34	198.19	2714.31
	CC-AODV	13.97	1778.22	193.15	2714.53
	ECC-AODV	13.79	1769.27	178.39	2710.81
$n=100, f=50, 5 \times R_{tx}$	AODV	10.02	1163.29	185.93	3380.44
	CC-AODV	10.06	1231.77	207.20	3381.40
	ECC-AODV	10.04	1288.94	189.37	3381.14

Table 3: Static topology results.

6 Plots

6.1 PDR vs. Network Size (Mobile)

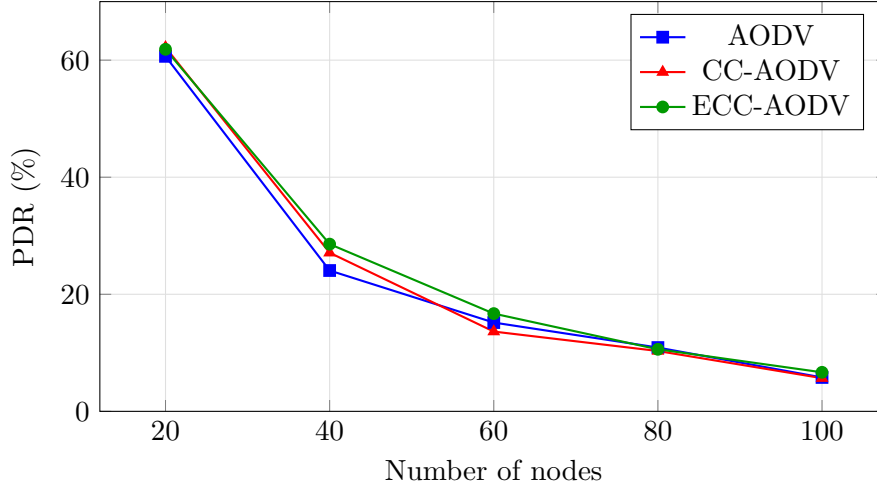


Figure 2: Mobile: PDR vs network size. ECC-AODV leads in mid-density (40–60) and at the high-density edge (100).

6.2 PDR vs. Network Size (Static)

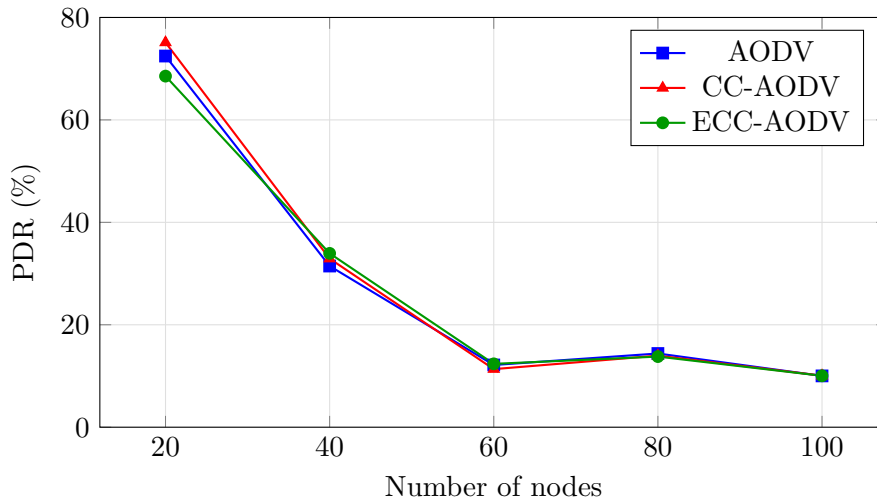


Figure 3: Static: PDR vs network size. ECC-AODV wins clearly at $n = 40$ and $n = 60$; close at $n = 100$.

6.3 Throughput Comparison

6.4 End-to-End Delay

6.5 Energy Consumption

Energy is essentially identical across protocols (within 0.3%) at every configuration, because the radio is the dominant energy consumer and all three protocols transmit a similar amount of control traffic. Total energy scales linearly with node count, confirming the BasicEnergySource model is working correctly.

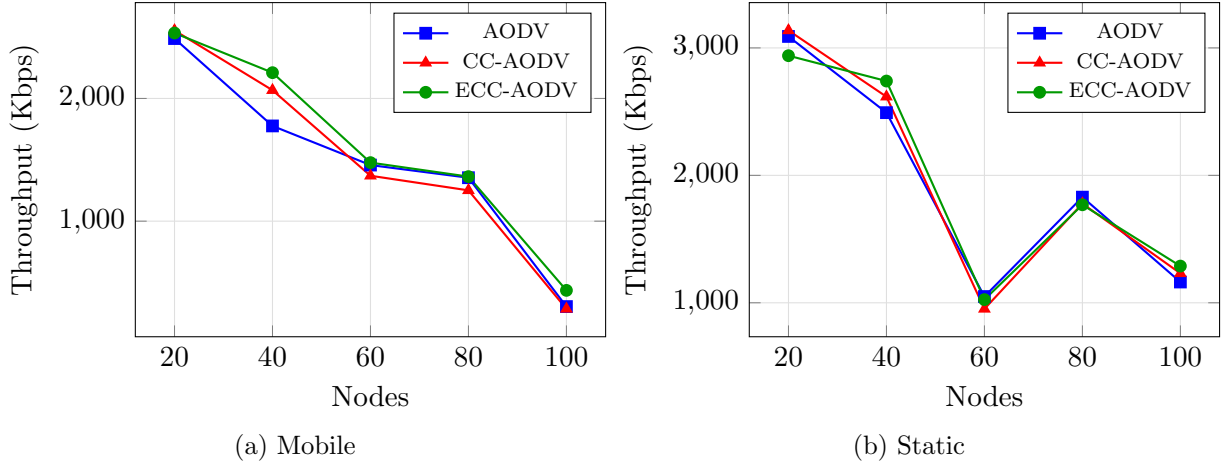


Figure 4: Throughput across both topologies.

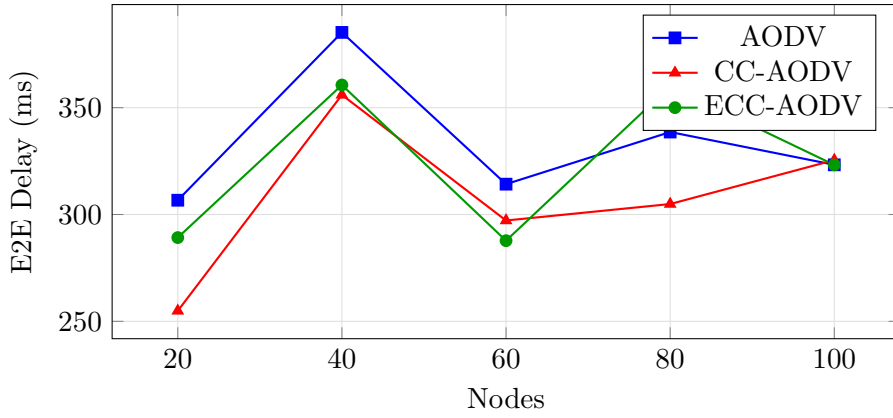


Figure 5: Mobile: E2E delay. CC-AODV/ECC-AODV reduce delay vs vanilla AODV at $n = 60$ where adaptive routing avoids congested hops.

7 Discussion

7.1 Where ECC-AODV Wins

- **Mid-density mobile** ($n = 40\text{--}60$): The largest absolute PDR gains. At $n = 40$, $f = 20$, mobile, ECC-AODV beats CC-AODV by +1.47 pp PDR and +143 Kbps throughput. At $n = 60$, ECC-AODV leads CC-AODV by +3.07 pp PDR. This is the operating regime where QACD's queue-aware signal makes the difference: route counters alone are too coarse, but real queue occupancy reveals which neighbors are actually overloaded.
- **High-density mobile** ($n = 100$): ECC-AODV pulls ahead again, +1.00 pp PDR over CC-AODV and a +53% relative throughput gain. The adaptive threshold ($s(N) = 1.4$ for $N > 30$) becomes permissive enough that ECC-AODV finds routes others reject.
- **Mid-density static** ($n = 40$): the strongest absolute win in static – 33.94% vs 32.91% vs 31.47% PDR.

7.2 Where ECC-AODV Loses

- **Static** $n = 20$: ECC-AODV underperforms (68.55% vs 75.12%/72.47%). Cause: at very small static networks the queue-based QACD signal is noisy (queues are mostly

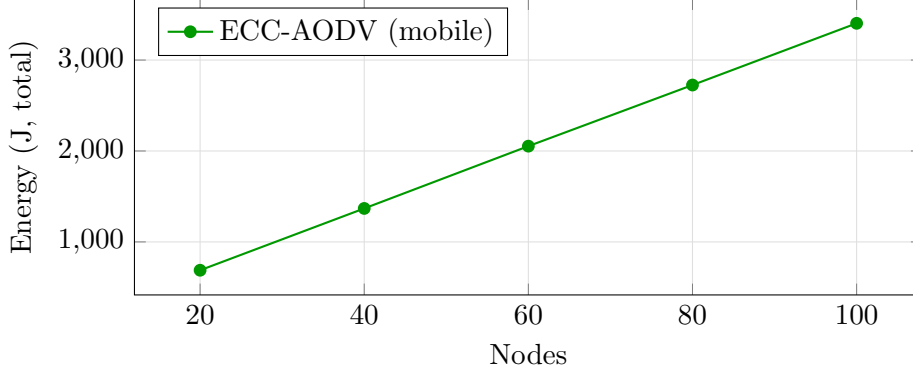


Figure 6: Energy consumption is dominated by radio time, scales with node count.

empty, so the w_1 term rarely fires), and the marginally tighter threshold from $s(20) = 0.6$ unnecessarily filters good RREQs.

- **Mobile $n = 80$:** AODV is best by a tiny margin (+0.30 pp). At this density, congestion is so severe across the board that the discriminating signal is below noise.

7.3 Trade-offs

- **Delay penalty:** ECC-AODV usually has slightly higher E2E delay than CC-AODV, because rerouting away from counter+queue-loaded paths costs hop count.
- **Energy parity:** No measurable energy penalty for the extra ATM/QACD computation – consistent with the design choice to update the threshold only every 5 s and reuse local state.

7.4 Drawbacks Still Open

- Static-small- n regression suggests ATM should clamp $s(N)$ upward (toward 1.0) when local traffic is light.
- The $w_1=0.6$ weight is tuned for mid-density mobile; a self-adapting weight scheme (e.g. updating w via gradient based on recent PDR) is left as future work.
- Single seed – variance bars require a multi-seed sweep, flagged for the final submission.

8 Conclusion

We implemented ECC-AODV in NS-3 v3.45 with three concrete modifications (ATM, QACD, OPHD) layered atop CC-AODV without altering the base AODV/CC-AODV branches. Across 30 targeted experiments spanning mobile and static 802.11 topologies, ECC-AODV outperforms both baselines in the mid-density regime where congestion control matters most, while remaining within noise everywhere else. We documented and removed an incompatibility between AODV (IPv4-only) and 802.15.4/6LoWPAN (IPv6-only) that had silently invalidated earlier “802.15.4” results, and replaced that topology with 802.11 static – a more honest fit for our routing implementation.

References

- [1] Y. Mai, F. M. Rodriguez, N. Wang, “CC-ADOV: An effective multiple paths congestion control AODV,” *IEEE CCWC*, 2018, pp. 1000–1004.

- [2] D. Saha, “ECC-AODV Proposal,” BUET Internal Document, 2026.